# EXPERIMENT-1

## Basic Networking Commands in Linux and Windows operating systems

**Aim:** To study and execute basic networking commands in Windows and Linux operating systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status, and troubleshooting network-related issues.

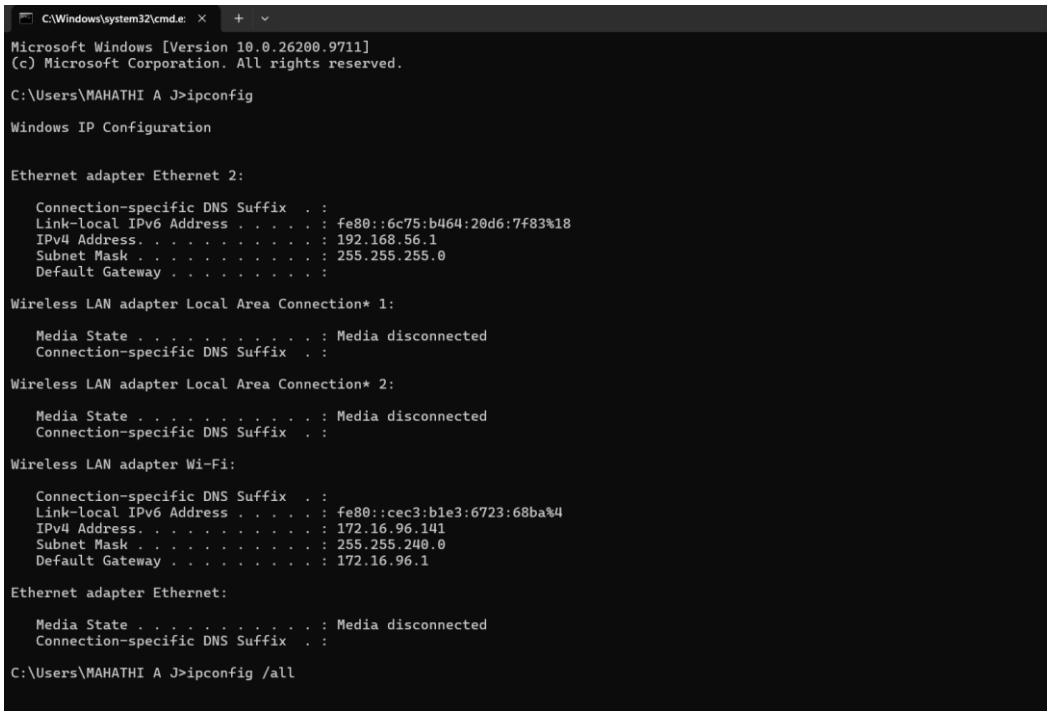
**Software Requirements :**

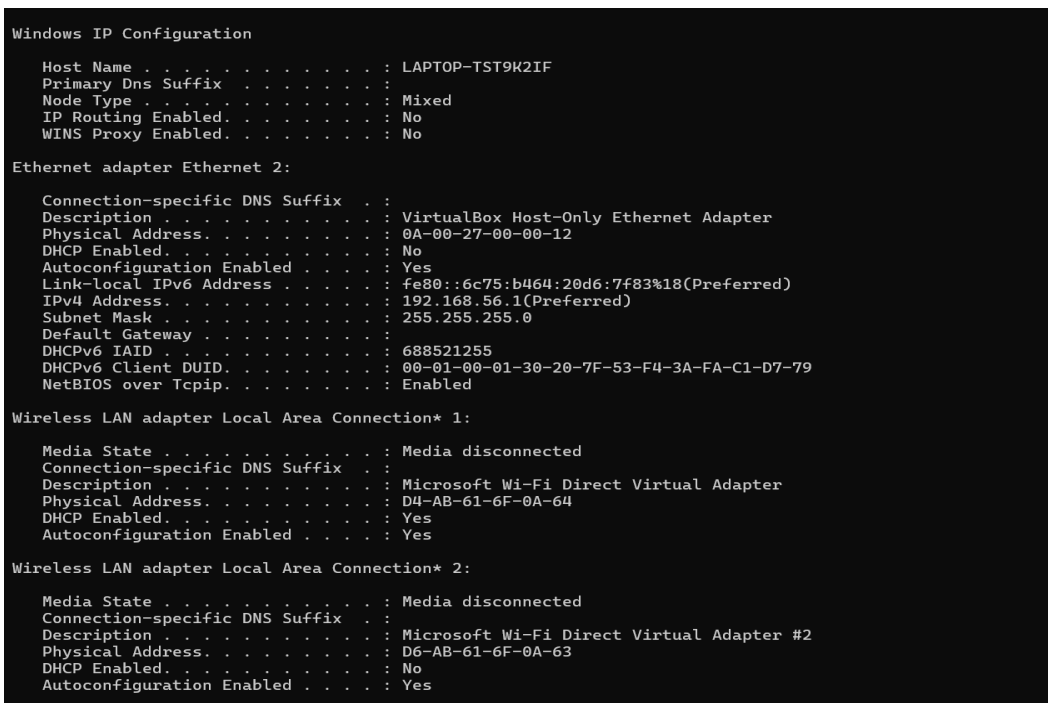
• Windows 10/11 or Linux (Ubuntu 20.04/22.04)

• Command Prompt (Windows)

• Terminal (Linux)

• Active Network Connection (LAN/Wi-Fi)



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Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX203
   Physical Address. . . . . . . . . : D4-AB-61-6F-0A-63
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::cec3:b1e3:6723:68ba%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.16.96.141(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Lease Obtained. . . . . . . . . . : 22 October 2026 08:14:31
   Lease Expires . . . . . . . . . . : 22 October 2026 08:29:30
   Default Gateway . . . . . . . . . : 172.16.96.1
   DHCP Server . . . . . . . . . . . : 172.16.96.1
   DHCPv6 IAID . . . . . . . . . . . : 81046369
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-20-7F-53-F4-3A-FA-C1-D7-79
   DNS Servers . . . . . . . . . . . : 172.16.96.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : F4-3A-FA-C1-D7-79
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\MAHATHI A J>ping google.com

Pinging google.com [142.250.67.46] with 32 bytes of data:
Reply from 142.250.67.46: bytes=32 time=9ms TTL=119
Reply from 142.250.67.46: bytes=32 time=13ms TTL=119
Reply from 142.250.67.46: bytes=32 time=18ms TTL=119
Reply from 142.250.67.46: bytes=32 time=4ms TTL=119

Ping statistics for 142.250.67.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 18ms, Average = 11ms
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C:\Users\MAHATHI A J>tracert google.com

Tracing route to google.com [142.250.67.46]
over a maximum of 30 hops:

  1     3 ms     9 ms     4 ms  172.16.96.1
  2   331 ms   101 ms    13 ms  61.95.128.185
  3    22 ms   702 ms     7 ms  116.119.36.60
  4     *        *        *     Request timed out.
  5     5 ms     6 ms    16 ms  pnmaaa-bb-in-f14.1e100.net [142.250.67.46]

Trace complete.

C:\Users\MAHATHI A J>nslookup google.com
Server:  UnKnown
Address:  172.16.96.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:805::200e
          142.250.67.46


C:\Users\MAHATHI A J>route print
===========================================================================
Interface List
 18...0a 00 27 00 00 12 ......VirtualBox Host-Only Ethernet Adapter
  3...d4 ab 61 6f 0a 64 ......Microsoft Wi-Fi Direct Virtual Adapter
 10...d6 ab 61 6f 0a 63 ......Microsoft Wi-Fi Direct Virtual Adapter #2
  4...d4 ab 61 6f 0a 63 ......Intel(R) Wi-Fi 6 AX203
 11...f4 3a fa c1 d7 79 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.16.96.1    172.16.96.141     30
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
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```
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      172.16.96.0    255.255.240.0         On-link      172.16.96.141    286
    172.16.96.141  255.255.255.255         On-link      172.16.96.141    286
   172.16.111.255  255.255.255.255         On-link      172.16.96.141    286
     192.168.56.0    255.255.255.0         On-link       192.168.56.1    281
     192.168.56.1  255.255.255.255         On-link       192.168.56.1    281
   192.168.56.255  255.255.255.255         On-link       192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link       192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link      172.16.96.141    286
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link       192.168.56.1    281
  255.255.255.255  255.255.255.255         On-link      172.16.96.141    286
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 18    281 fe80::/64                On-link
  4    286 fe80::/64                On-link
 18    281 fe80::6c75:b464:20d6:7f83/128
                                    On-link
  4    286 fe80::cec3:b1e3:6723:68ba/128
                                    On-link
  1    331 ff00::/8                 On-link
 18    281 ff00::/8                 On-link
  4    286 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None

C:\Users\MAHATHI A J>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        LAPTOP-TST9K2IF:49676  ESTABLISHED
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```
  TCP    127.0.0.1:49676        LAPTOP-TST9K2IF:49675  ESTABLISHED
  TCP    172.16.96.141:51482    104.18.32.47:https     ESTABLISHED
  TCP    172.16.96.141:53398    4.213.25.242:https     ESTABLISHED
  TCP    172.16.96.141:53471    52.123.128.14:https    CLOSE_WAIT
  TCP    172.16.96.141:53502    24:https               TIME_WAIT
  TCP    172.16.96.141:53503    52.168.117.173:https   TIME_WAIT
  TCP    172.16.96.141:53505    74.226.60.120:https    CLOSE_WAIT
  TCP    172.16.96.141:53516    4.150.223.105:https    TIME_WAIT
  TCP    172.16.96.141:53520    161.69.232.96:https    TIME_WAIT
  TCP    172.16.96.141:53547    20.184.175.22:https    TIME_WAIT
  TCP    172.16.96.141:55092    sb-in-f188:5228        ESTABLISHED
  TCP    172.16.96.141:58556    104.18.32.47:https     ESTABLISHED
  TCP    172.16.96.141:58701    1:https                ESTABLISHED
  TCP    172.16.96.141:58708    150.171.110.87:https   ESTABLISHED
  TCP    172.16.96.141:58709    150.171.110.87:https   ESTABLISHED
  TCP    172.16.96.141:58710    a23-201-53-136:https   ESTABLISHED
  TCP    172.16.96.141:58711    a23-201-53-136:https   ESTABLISHED
  TCP    172.16.96.141:58712    52.123.253.103:https   ESTABLISHED
  TCP    172.16.96.141:58713    52.123.253.103:https   TIME_WAIT
  TCP    172.16.96.141:58714    4.150.223.105:https    ESTABLISHED
  TCP    172.16.96.141:58716    172.64.155.209:https   TIME_WAIT
  TCP    172.16.96.141:59125    sb-in-f188:5228        ESTABLISHED
  TCP    172.16.96.141:59303    104.18.37.228:https    ESTABLISHED
  TCP    172.16.96.141:61827    sb-in-f188:5228        ESTABLISHED
  TCP    172.16.96.141:62183    104.18.42.202:https    CLOSE_WAIT
  TCP    172.16.96.141:62316    104.21.76.26:https     ESTABLISHED
  TCP    172.16.96.141:62317    104.18.36.61:https     CLOSE_WAIT
  TCP    172.16.96.141:62366    4.213.25.242:https     ESTABLISHED
  TCP    172.16.96.141:63068    pnmaaa-az-in-f14:https  CLOSE_WAIT

C:\Users\MAHATHI A J>
C:\Users\MAHATHI A J>netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:445            LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:3306           LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:5040           LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:33060          LAPTOP-TST9K2IF:0      LISTENING
```

```
C:\Users\MAHATHI A J>getmac

Physical Address    Transport Name
=================== ==========================================================
F4-3A-FA-C1-D7-79   Media disconnected
D4-AB-61-6F-0A-63   \Device\Tcpip_{1EA7D478-B256-4675-96D7-2F7B0F11B905}
0A-00-27-00-00-12   \Device\Tcpip_{F4A63484-1407-4F78-A027-5481FC323255}

C:\Users\MAHATHI A J>hostname
LAPTOP-TST9K2IF

C:\Users\MAHATHI A J>arp -a

Interface: 172.16.96.141 --- 0x4
  Internet Address      Physical Address      Type
  172.16.96.1           7c-5a-1c-cf-be-48     dynamic
  172.16.97.225         14-d4-24-19-8b-67     dynamic
  172.16.102.216        ec-3a-56-ce-a7-b8     dynamic
  172.16.111.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x12
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

C:\Users\MAHATHI A J>
```